

Computer Networks - Module 5 Notes

Application Layer

1. Introduction to Application Layer:

- The topmost layer of the network stack.
- Provides services directly to the end user.
- Interfaces with software applications to implement communication over a network.
- Examples: Email, File Transfer, Web browsing, DNS, Remote login.

2. Domain Name System (DNS):

- Hierarchical distributed naming system for computers and services.
- Converts human-readable domain names (www.example.com) into IP addresses.
- Components:
 - * Domain Names: Structured in hierarchy (root, TLD, subdomain).
 - * Name Servers:
 - Root Servers (top-level).
 - TLD Servers (.com, .org, .in).
 - Authoritative Servers (store domain info).
 - * Resolvers: Client-side software that initiates queries.
- DNS Record Types: A (address), AAAA (IPv6), MX (mail), CNAME (alias).

3. Simple Mail Transfer Protocol (SMTP):

- Protocol for sending email between servers (push protocol).
- Uses TCP port 25 (or 587 for submission).
- Process:
 - * User Agent composes message.
 - * Message sent to SMTP server.
 - * Server relays message to recipient's mail server.
- For receiving mail: POP3 (Post Office Protocol) and IMAP are used.

4. File Transfer Protocol (FTP):

- Used to transfer files between client and server.
- Operates over TCP, uses two connections:
 - * Control Connection (port 21).
 - * Data Connection (port 20).
- Supports authentication (username, password).
- Modes: Active & Passive.

5. World Wide Web (WWW):

- A system of interlinked hypertext documents accessed via the Internet.
- Components:
 - * Web Servers: Store web pages.
 - * Web Clients (Browsers): Request & render pages.
 - * Hyperlinks: Connect documents.
- Uses HTTP/HTTPS for communication.

6. Hypertext Transfer Protocol (HTTP):

- Application layer protocol for communication between client and server.
- Stateless protocol (each request independent).
- Uses TCP port 80 (HTTPS uses port 443 with SSL/TLS).